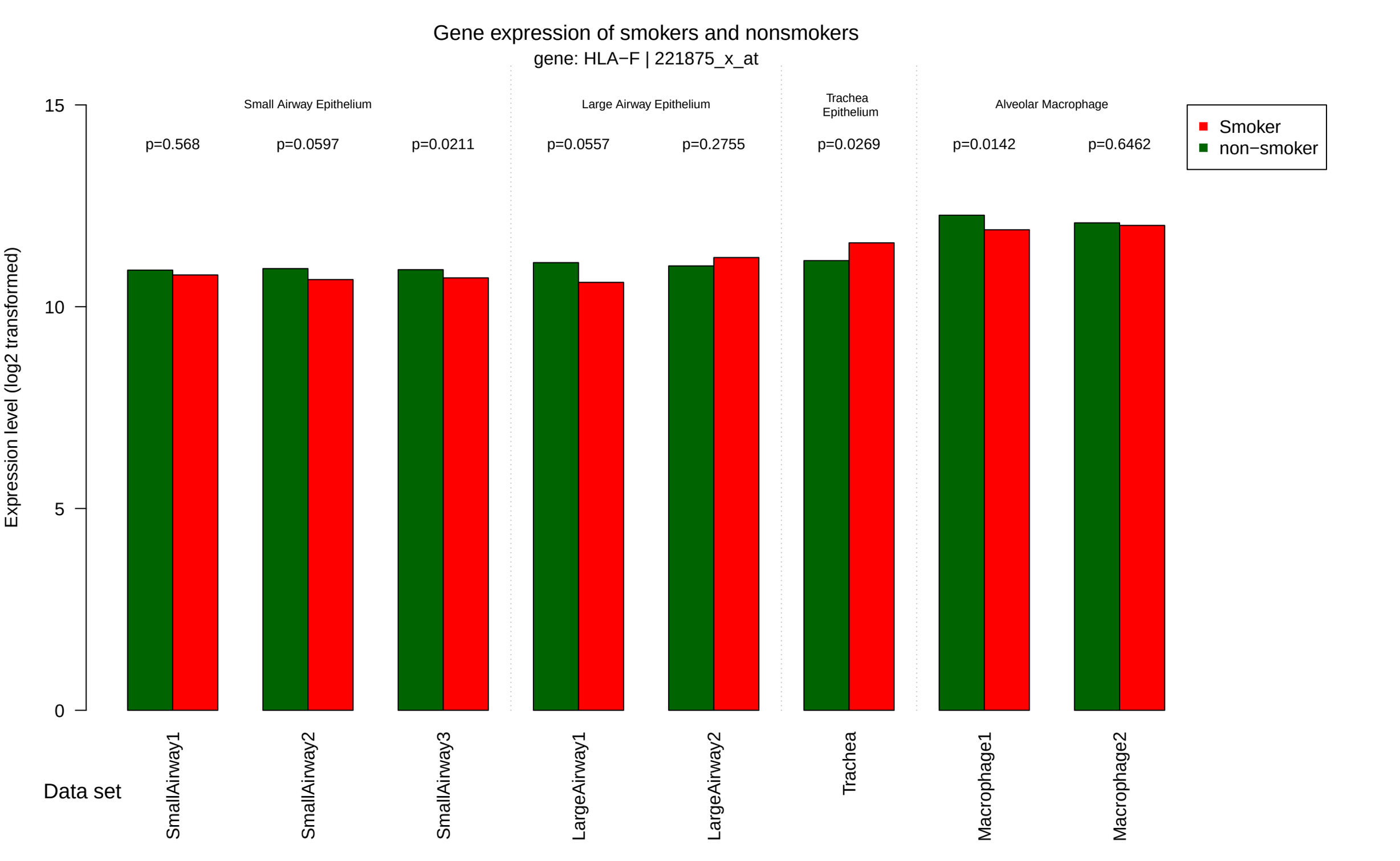
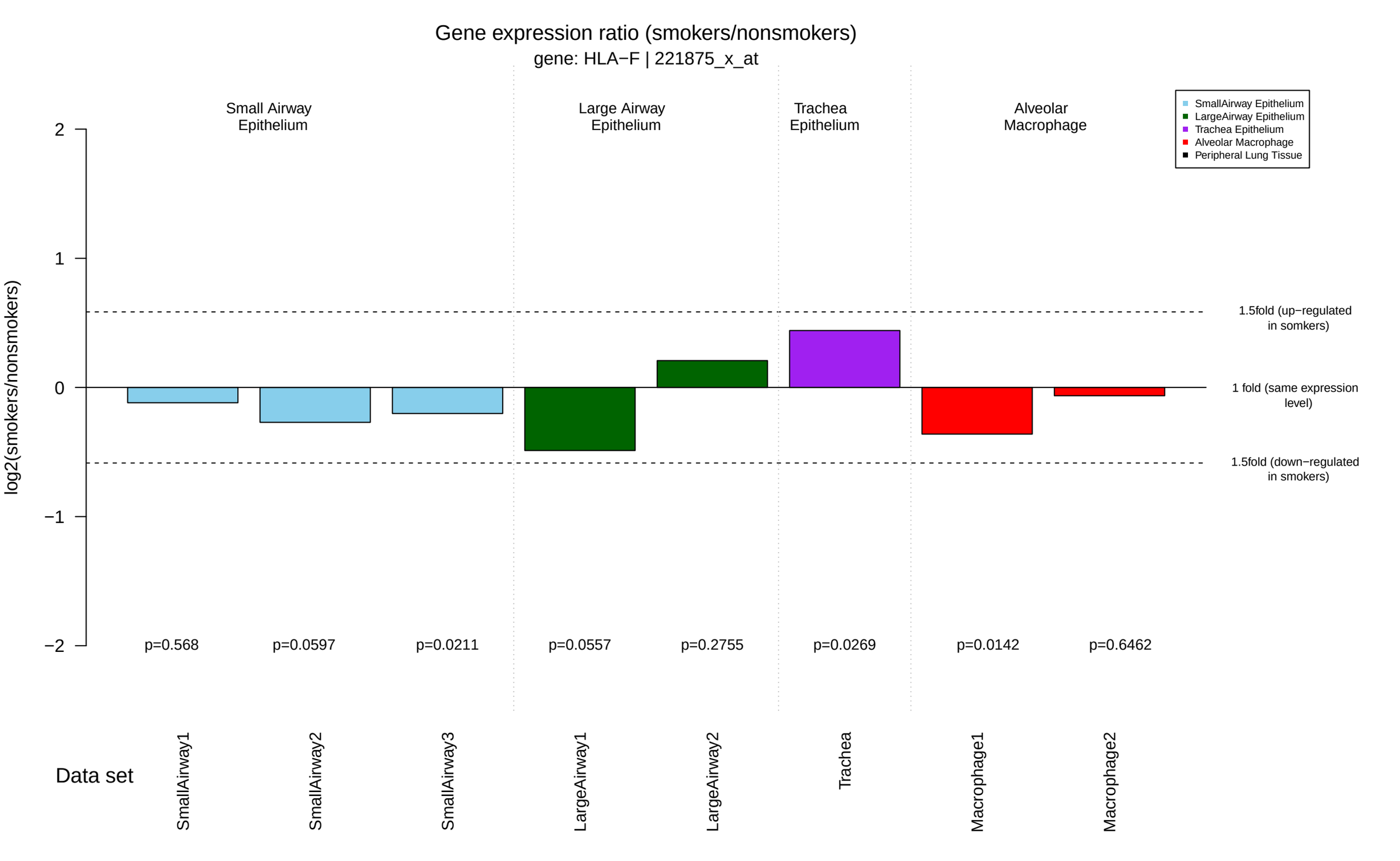
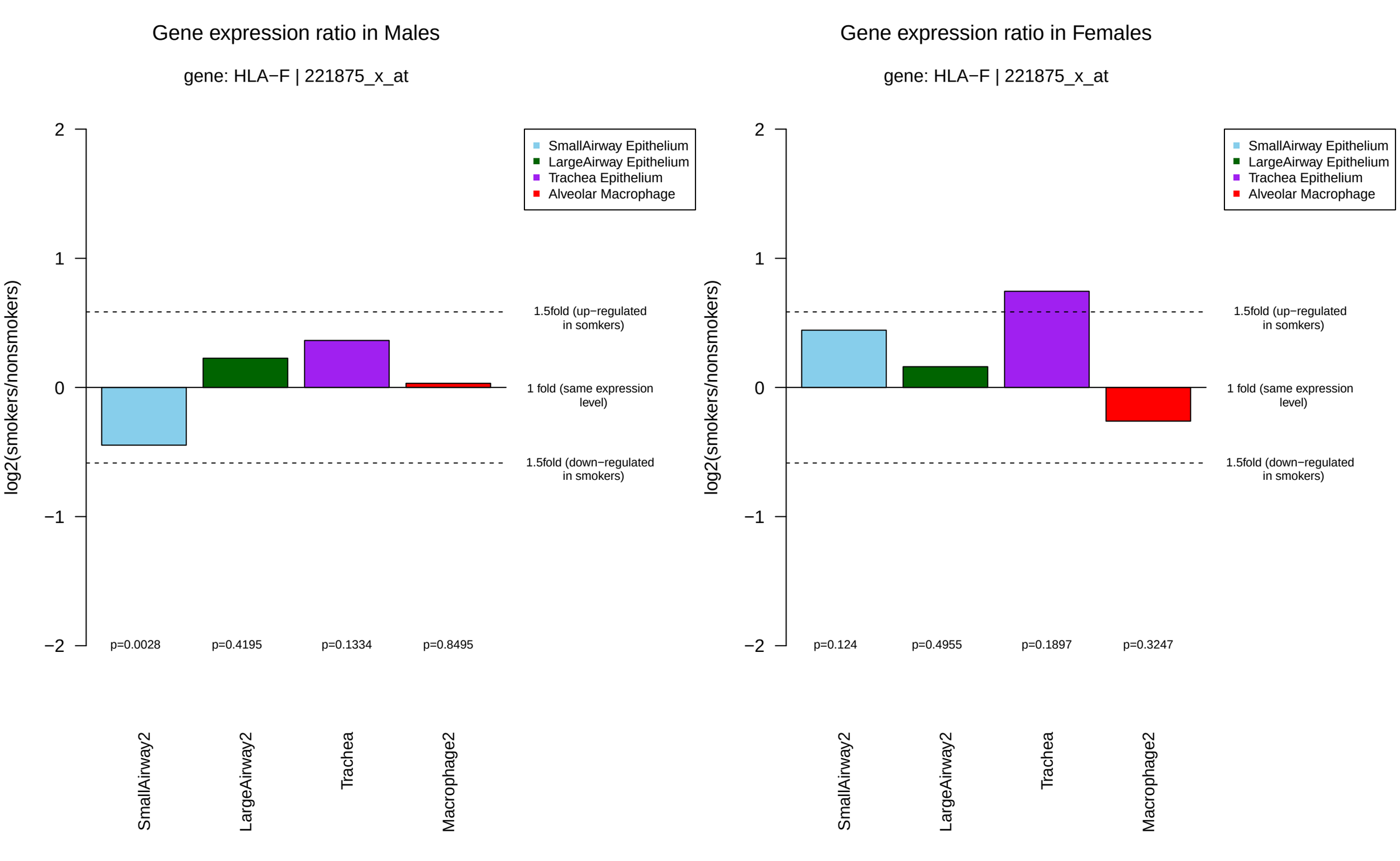
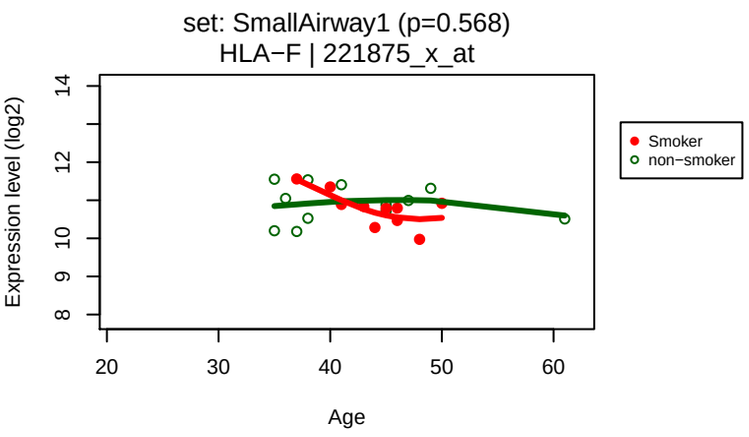
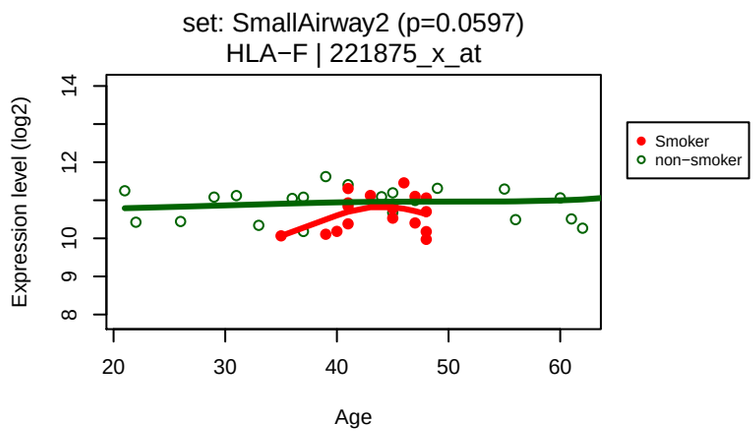
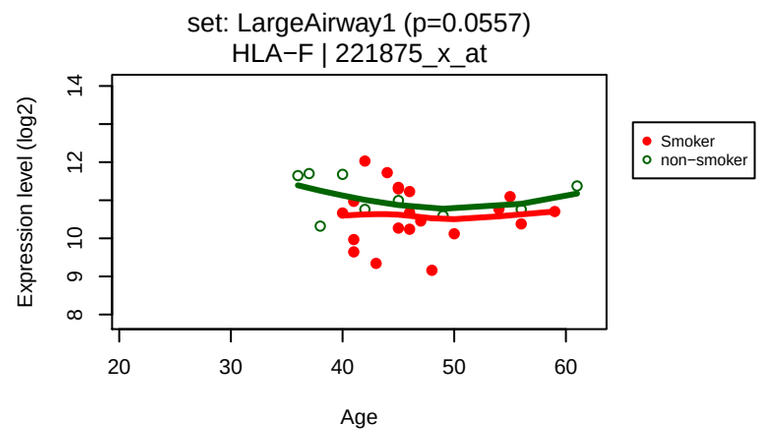

Gene expression of smokers and nonsmokers
gene: HLA−F | 221875_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.568
p=0.0597
p=0.0211
p=0.0557
p=0.2755
p=0.0269
p=0.0142
p=0.6462
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

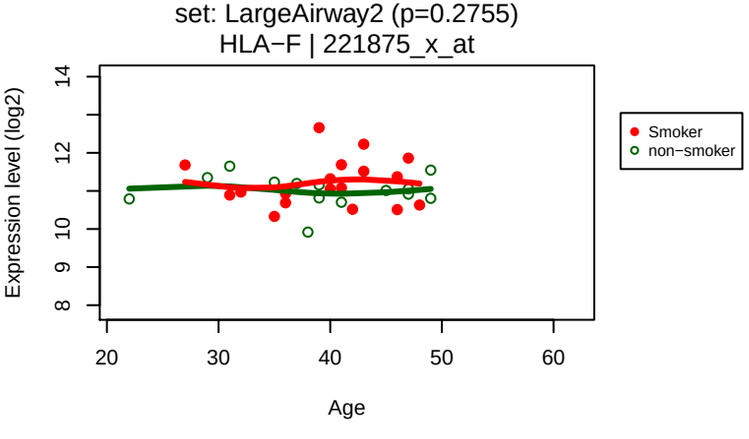

Gene expression ratio (smokers/nonsmokers)
gene: HLA−F | 221875_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.568
p=0.0597
p=0.0211
p=0.0557
p=0.2755
p=0.0269
p=0.0142
p=0.6462
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

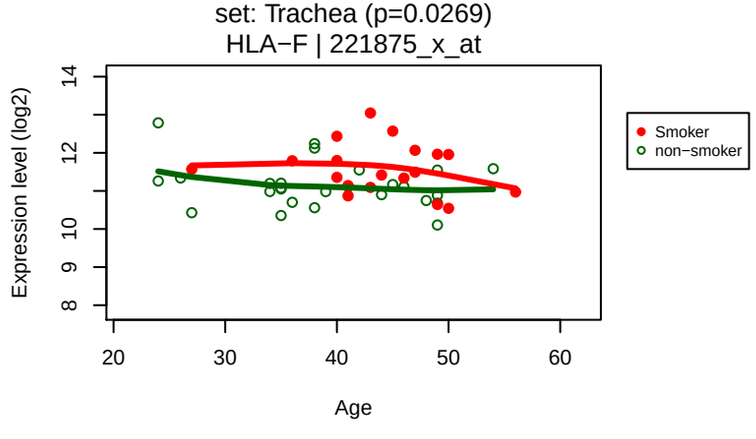

Gene expression ratio in Males
gene: HLA−F | 221875_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0028
p=0.4195
p=0.1334
p=0.8495
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−F | 221875_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.124
p=0.4955
p=0.1897
p=0.3247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

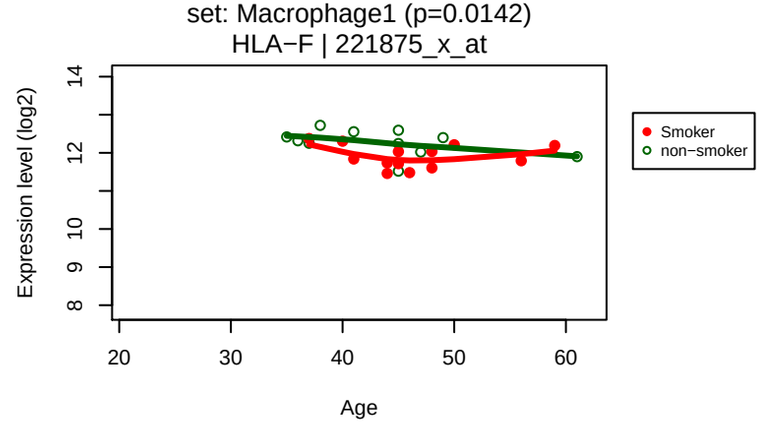

set: SmallAirway1 (p=0.568)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker

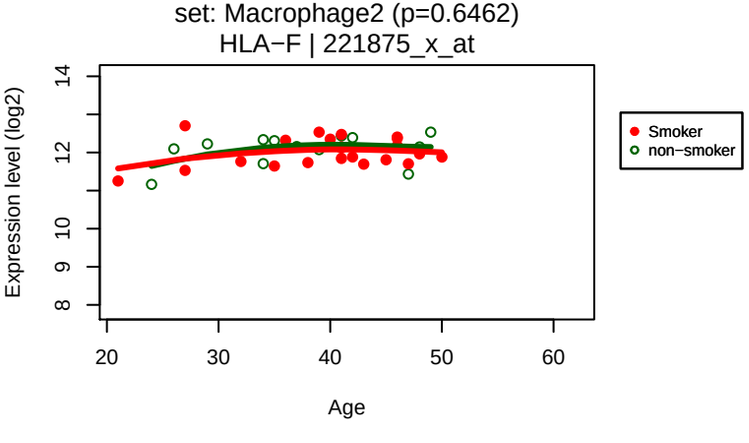

set: SmallAirway2 (p=0.0597)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker

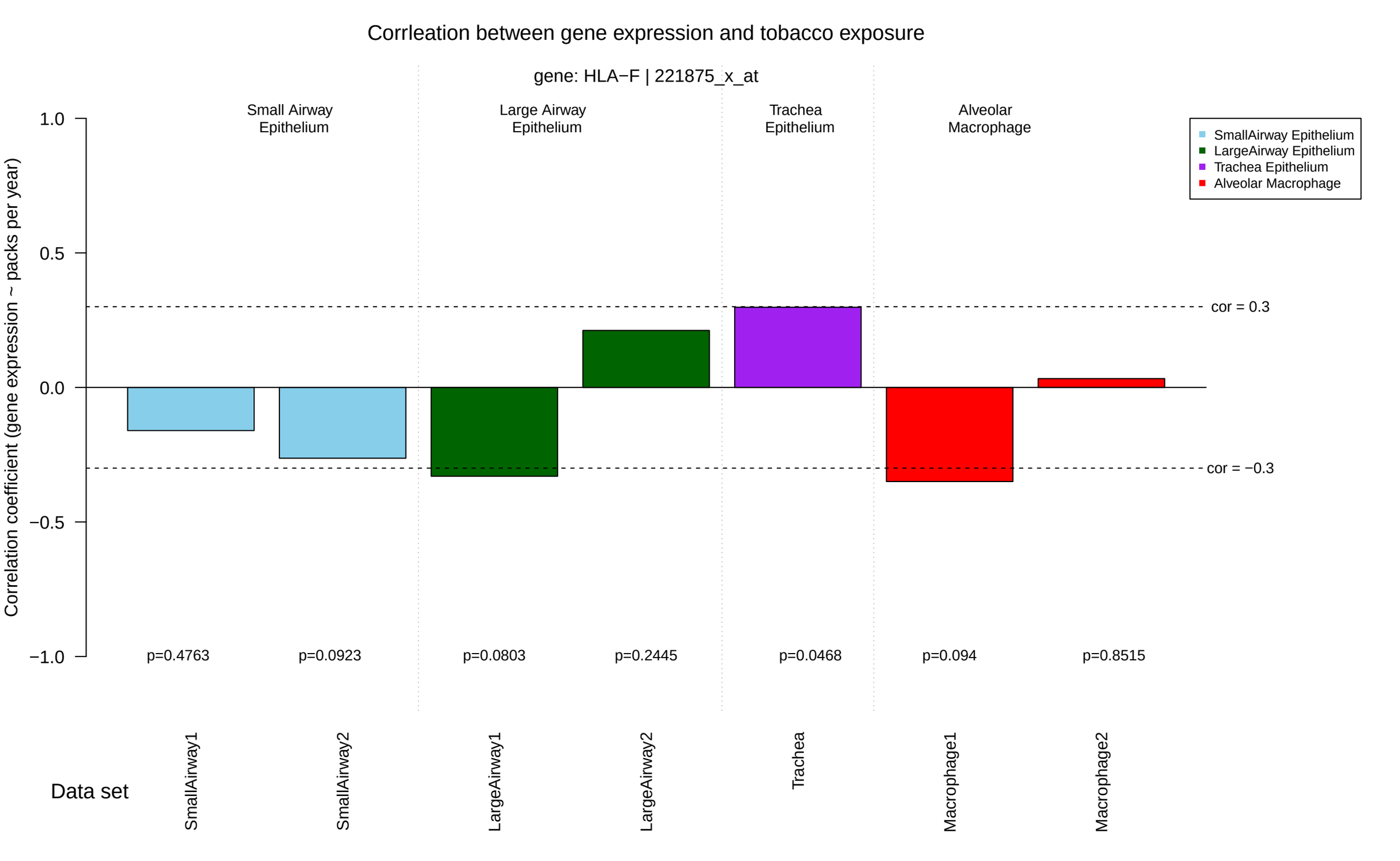

set: LargeAirway1 (p=0.0557)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2755)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0269)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0142)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6462)
HLA−F | 221875_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−F | 221875_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4763
p=0.0923
p=0.0803
p=0.2445
p=0.0468
p=0.094
p=0.8515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2